

Aniket Ghosh

ghosh.anik@northeastern.edu • Boston, MA • Portfolio • LinkedIn • GitHub • LessWrong

Education

M.S. in Artificial Intelligence, ML Concentration – Northeastern University – GPA: 4.0/4.0 Aug 2025 – May 2027
Courses: Actionable Interpretable Methods, Reinforcement Learning, Machine Learning Operations (MLOps), Algorithms.

B.Tech in CS & Business Systems – Institute of Engineering & Management, Kolkata – GPA: 4.0/4.0 2020 – 2024
Ranked **2nd of 180+** • **Director's Award** • *Key Courses:* Neural Networks, NLP, Data Structures, OS.

Technical Skills

Languages: Python, C/C++, SQL, Bash

ML & Deep Learning: PyTorch, Hugging Face Transformers, scikit-learn, TensorFlow, NumPy, pandas

Training & Inference: distributed multi-GPU training (SLURM), LLM fine-tuning, LoRA/QLoRA, quantization

GenAI: RAG, LangChain, LangGraph, OpenAI Agents SDK, MCP, vector databases (ChromaDB)

MLOps: Docker, Kubernetes, AWS, GCP, CI/CD, GitHub Actions, MLflow, FastAPI, Terraform, Git, Linux

Domains: mechanistic interpretability, AI safety, NLP, computer vision, reinforcement learning

Experience

AI/ML Engineer – Varosync, Inc., San Francisco, CA (remote) May 2026 – Aug 2026

- Only ML engineer at an early-stage drug-discovery company, reporting to the CEO and CTO. Built the stack from scratch: a knowledge graph over millions of molecules, then featurization, vector indexing, retrieval and the live query path, where the probe-guided precision method below went into production.
- Trained **7–8B parameter** molecular embedding models on an **800M-molecule** corpus across a multi-GPU cluster, owning the whole loop from data curation and distributed training through checkpointing and evaluation.
- Caught the retrieval benchmark training on future outcomes in **123 of 133** cases. Swapped k-fold for a leave-one-group-out, point-in-time split and took post-cutoff training evidence from **21.8% to 0%**, losing one case to do it.

AI Researcher – CMATER Lab, Jadavpur University, Kolkata Jun 2023 – May 2025

- Built an unsupervised ML pipeline for histopathological cell segmentation using **color quantization** and **DBSCAN clustering**, achieving **85%+ accuracy** and **30% fewer false positives** vs. a Mask R-CNN baseline — enabling faster, annotation-free pathology screening.

Computer Vision Research Intern – North-Eastern Hill University (Remote) Feb 2023 – Jul 2023

- Conducted a **50+ paper literature review** on sign language recognition to define annotation and quality standards for a novel **Indian Sign Language benchmark dataset**, adapting best practices from Microsoft ASL and MNIST to underrepresented language contexts.

Projects

Probe-Guided Mixed-Precision Quantization LLM Compression • LessWrong • GitHub

- Trained a linear probe at every layer to find where NER, POS and chunking signal gets built, then protected those layers and compressed the rest hard. At a **5-bit average** the map held **99–100% of full-precision accuracy** on three unseen datasets. Spending the same budget evenly kept **16–41%**.
- Went into production at Varosync, deciding where capacity goes across layers in the 7–8B molecular embedding models.

Biomedical Knowledge Graph Link Prediction Healthcare AI • GitHub

- Engineered graph features (PageRank, structural metrics) on the **BioRED** biomedical corpus to train a **Random Forest** link prediction model achieving **0.94 ROC-AUC** — a **23%+ gain** over learned embeddings — and implemented **5-hop reasoning** for relationship explainability.
- Ran a rigorous **classical ML vs. TransE KG embedding** benchmark (0.94 vs. 0.61 ROC-AUC), demonstrating that hand-crafted graph features dominate learned representations on low-resource biomedical corpora — a practical finding for data-scarce production settings.

Autonomous Multi-Agent Trading Simulation GitHub

- Designed and deployed a **multi-agent trading simulation** using the **OpenAI Agents SDK: 4 autonomous AI traders** coordinated across **6 MCP servers** with **44 distinct tools**, consuming live market data via **Polygon.io** for end-to-end autonomous portfolio management.

Publications & Patents

Intelligent Traffic Sign Detection Using YOLOv8 with Hybrid CNN Filtering *First Author*

Published – AISC 2024 • Springer “Information and Communication Technologies” series

Lightweight Hybrid DNN-GNN for Network Intrusion Detection with Adaptive Late Fusion *Co-author*

Published – IEEE • Patent Filed by SRM Institute of Science and Technology

Achievements & Leadership

- **BlueDot Impact Technical AI Safety** — certified July 2026
- **Teaching Co-Lead, AIDE Program:** Designing a **12-session ML lab** for philosophy PhD students at Northeastern's AI and Data Ethics Summer Program (Jun–Aug 2026).
- **Graduate TA:** Lead PyTorch and deep learning labs for **40+ graduate students** in NLP at Northeastern University.
- **Academic Excellence:** Ranked **2nd out of 180+ students** in B.Tech; awarded the **Director's Award**.
- **Volunteer Educator:** **7 years** teaching math & CS to **underprivileged students** at The Hope Foundation.
- **Graduate Student Senator:** Elected to Northeastern's **Graduate Student Government**.